$V$ spațiu vectorial peste corpul $K$

$\mathcal{B} = \{v_1 \dots v_n\} \subseteq V$ bază $\Leftrightarrow$

a) $\forall \lambda_1 \dots \lambda_n \in K$ a.î. $\lambda_1 v_1 + \dots + \lambda_n v_n = 0 \Rightarrow$

$\lambda_1 = \dots = \lambda_n = 0$

b) $\forall v \in V \; \exists \lambda_1 \dots \lambda_n \in K$ a.î.

$v = \lambda_1 v_1 + \dots + \lambda_n v_n$

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ex.

$$v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = x_1 \underbrace{\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{e_1} + \dots + x_n \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}}_{e_n} \qquad \forall v \in K^n$$

$\Rightarrow$ $e_1 \dots e_n$ sistem de generatori pt $K^n$

$Sp\{e_1 \dots e_n\} = K^n$

$e_1 \dots e_n$ liniar indep? Fie $\lambda_1 \dots \lambda_n \in K$ a.î.

$$\begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \underbrace{\lambda_1 e_1 + \dots + \lambda_n e_n} = 0_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \Rightarrow \lambda_1 = \dots = \lambda_n = 0$$

$\Rightarrow$ $\mathcal{B}_c = \{e_1, \dots e_n\}$ bază în $k^n$

bază canonică

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ex. $V = k[X]$ $\quad \mathcal{B} = \{1, X, X^2, \dots, X^n, \dots\}$

bază în $V$

/ lin. indep.

fie $X^{n_1}, \dots, X^{n_p} \in \mathcal{B}$

$n_i \neq n_j$, $i, j = \overline{1,p}$

pp $n_1 < n_2 < \dots < n_p$.

fie $\lambda_1 X^{n_1} + \dots + \lambda_p X^{n_p} = 0$.

$\Leftrightarrow \lambda_1 = \dots = \lambda_p = 0$

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$f : \mathbb{R} \longrightarrow \mathbb{R}$

$f(x) = \lambda_1 x^{n_1} + \dots + \lambda_p x^{n_p}$

$f = 0 \quad \Leftrightarrow \quad f(x) = 0 \quad \forall x \in \mathbb{R}$

$\Rightarrow \lambda_1 = \dots = \lambda_p = 0$ ?

$f(x) = 0 \quad / \ (-)^{(n_1)}$

$\lambda_1 n_1! + \lambda_2 \frac{n_2!}{(n_2 - n_1 + 1)!} x^{n_2 - n_1} + \dots = 0$

$\begin{vmatrix} & & \lambda_p n_p! \\ 0 & 0 & \\ & & \end{vmatrix} = 0$

$\Rightarrow \quad 1, x, x^2, \dots, x^n, \dots$ liniar indep.

$Sp\{1, x, \dots, x^n, \dots\} = k[x]$

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$V = k_n[x] = \{ P \in k[x] \mid \text{grad } P \leq n \}$

bază canonică $\mathcal{B} = \{ \underbrace{1, x, \dots, x^{n-1}, x^n}_{n+1} \}$

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$V = \mathcal{M}_2(\mathbb{R})$ $\quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$A = a\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$E_{11}$ $\quad E_{12}$ $\quad E_{21}$ $\quad E_{22}$

$\{E_{11}, E_{12}, E_{21}, E_{22}\}$ bază $\mathcal{M}_2(\mathbb{R})$

$2 \cdot 2 = 4$ matrice.

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$V$ bază $\begin{cases} \mathcal{B}_1 & n \text{ vectori} \quad v_1 \dots v_n \\ \mathcal{B}_2 & m \text{ vectori} \quad w_1 \dots w_m \end{cases}$

$\Rightarrow \quad m \leq n \quad , \quad n \leq m \quad \Rightarrow m = n$

$\dim_{\mathbb{K}} V = n$

**Prop.** $V$ sp. vect. peste $k$

$\mathcal{B} = \{v_1, \dots v_n\}$ bază în $V$

$w_1, \dots w_m \in V$ liniar indep. $\Rightarrow m \leq n$.

$P \Rightarrow Q$

$\neg Q \Rightarrow \neg P$.

**Dem.** Pp. că $m > n$.

Vrem să vedem că $w_1 \dots w_m$ liniar dep.

Fie $\alpha_{ij} \in k$ a.î.

$$w_i = \sum_{j=1}^{n} \alpha_{ij} v_j \quad \forall i = \overline{1, m}$$

Fie $\lambda_i \in k$, $i = \overline{1, m}$ a.î. $\sum_{i=1}^{m} \lambda_i w_i = 0$

$$\Rightarrow 0 = \sum_{i=1}^{m} \lambda_i w_i = \sum_{i=1}^{m} \lambda_i \sum_{j=1}^{n} \alpha_{ij} v_j =$$

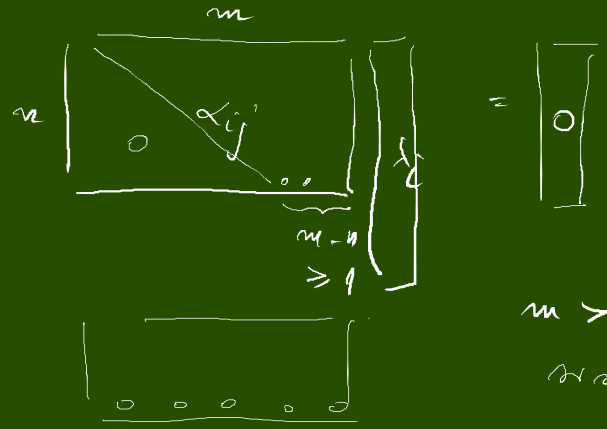
$$= \sum_{j=1}^{n} \underbrace{\left( \sum_{i=1}^{m} \lambda_i \alpha_{ij} \right)}_{k} \cdot \underbrace{v_j}_{V}$$

$\{v_1, \dots v_n\}$ bază $\Rightarrow v_1, \dots v_n$ liniar independenți.

$$\Rightarrow \sum_{i=1}^{m} \lambda_i \alpha_{ij} = 0 \quad \forall j = \overline{1, n}$$

omogen

Sistem de $n$ ec. cu $\lambda_1, \dots \lambda_m$ $m$ necunoscute




$m > n \Rightarrow$ sist. compat. nedet.

$\Rightarrow$ există $\lambda_1 \dots \lambda_m$ nenule

$\Rightarrow w_1, \dots w_m$ liniar dep.

Contrad.

Exemplu. $V = \mathbb{R}^2$ $k = \mathbb{R}$ $\dim_{\mathbb{R}} \mathbb{R}^2 = 2$

a) $\left\{\begin{pmatrix}-1\\-3\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}1\\2\end{pmatrix}\right\}$ NU

b) $\left\{\begin{pmatrix}2\\3\end{pmatrix}\right\}$ NU

c) $\left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}\right\}$ ? DA

d) $\left\{\begin{pmatrix}3/5\\4/5\end{pmatrix}, \begin{pmatrix}-4/5\\3/5\end{pmatrix}\right\}$ ? temă

c) — sist. liniar indep. ✓

$\lambda_1, \lambda_2 \in \mathbb{R}$

$\lambda_1 \begin{pmatrix}1\\2\end{pmatrix} + \lambda_2 \begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$

$\begin{cases}\lambda_1 + \lambda_2 = 0 \\ 2\lambda_1 + \lambda_2 = 0\end{cases} \Rightarrow \lambda_1 = 0$

$\lambda_2 = 0$.

Sist. de gen. ✓

$\forall\, v \in \mathbb{R}^2$ $v = \begin{pmatrix}x\\y\end{pmatrix}$

$\exists\, \alpha_1, \alpha_2 \in \mathbb{R}$ a.î.

$v = \alpha_1 \begin{pmatrix}1\\2\end{pmatrix} + \alpha_2 \begin{pmatrix}1\\1\end{pmatrix}$

$$\begin{cases} \alpha_1 + \alpha_2 = x \\ 2\alpha_1 + \alpha_2 = y \end{cases}$$

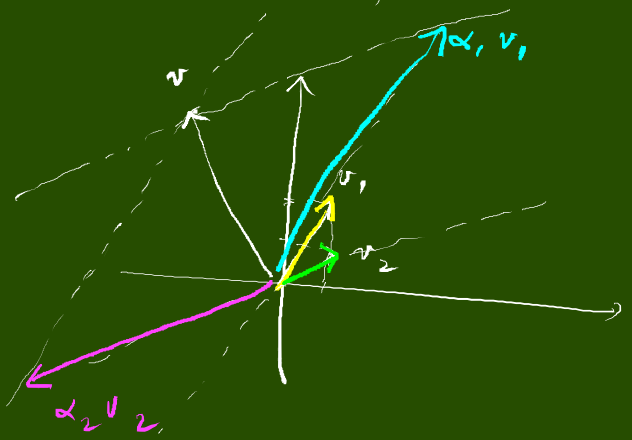
$\alpha_1 = y - x$

$\alpha_2 = x - \alpha_1 = 2x - y$

$\Rightarrow$ first. are sol.

$\Rightarrow \left\{ \underset{v_1}{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}, \underset{v_2}{\begin{pmatrix} 1 \\ 1 \end{pmatrix}} \right\}$ base in $\mathbb{R}^2$

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$




$\alpha_1 v_1 + \alpha_2 v_2 = v$

$v_1 \dots v_n \in V$ bază $\iff \forall v \in V \ \exists! \ \lambda_1 \dots \lambda_n \in k$ a.î. $v = \lambda_1 v_1 + \dots + \lambda_n v_n$

lin. indep. | Sist. de gen.

$\Rightarrow$ " temă

$\Leftarrow$ " Fie $\lambda_1 \dots \lambda_n \in k$ a.î.

$\lambda_1 v_1 + \dots + \lambda_n v_n = 0$

$0 \cdot v_1 + \dots + 0 \cdot v_n = 0$

Scrierea e unică $\Rightarrow \lambda_1 = \dots = \lambda_n = 0$

$\Rightarrow v_1 \dots v_n$ lin. indep.

**Corolar:** $v_1 \dots v_n \in k^n$ bază $\iff$

$\det(v_1 | \dots | v_n) \neq 0$.

$k^n$ $\mathcal{B}$ bază can. $\Rightarrow [v]_{\mathcal{B}} = v$

Exemples : $V_1 = \mathbb{R}^4$ $V_2 = M_2(\mathbb{R})$ $V_3 = \mathbb{R}_3[X]$

$\mathcal{B}_1$ $\mathcal{B}_2$ $\mathcal{B}_3$

$v = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 3 \end{pmatrix}$ $[v]_{\mathcal{B}_1} = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 3 \end{pmatrix}$

$v = 1 \cdot e_1 + (-2) e_2 + 0 \cdot e_3 + 3 \cdot e_4$

$A = \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$ $[A]_{\mathcal{B}_2} = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 3 \end{pmatrix}$

$A = 1 \cdot E_{11} + (-2) E_{12} + 0 \cdot E_{21} + 3 \cdot E_{22}$

$P(X) = 1 - 2X + 3X^3$ $[P(X)]_{\mathcal{B}_3} = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 3 \end{pmatrix}$

$P(X) = 1 \cdot 1 + (-2) \cdot X + 0 \cdot X^2 + 3 \cdot X^3$

$B = \{v_1 \dots v_n\}$ base

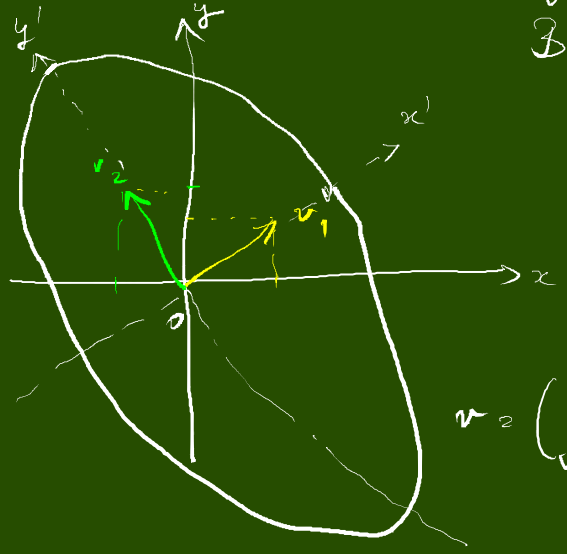
$\forall\, v \in V \quad \exists!\ \lambda_1 \dots \lambda_n \in k$

$v = \lambda_1 v_1 + \dots + \lambda_n v_n$

$v_1 = 1 \cdot v_1 + 0 \cdot v_2 + \dots + 0 \cdot v_n \Rightarrow [v_1]_B = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$

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$\{(x,y) \mid 18x^2 + 24xy + 11y^2 = 50\}$ ?




$B = \left\{ v_1 = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}, v_2 = \begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix} \right\}$

$v = \begin{pmatrix} x \\ y \end{pmatrix} \qquad [v]_B = \begin{pmatrix} x' \\ y' \end{pmatrix}$

$x' = ? \quad y' = ?$

$v = \begin{pmatrix} x \\ y \end{pmatrix} = x' \cdot v_1 + y' \cdot v_2$

$$\begin{cases} x = \frac{4}{5}x' - \frac{3}{5}y' \\ y = \frac{3}{5}x' + \frac{4}{5}y' \end{cases}$$

$$\Rightarrow 18\left(\frac{4}{5}x' - \frac{3}{5}y'\right)^2 + 24\left(\frac{4}{5}x' - \frac{3}{5}y'\right)\left(\frac{3}{5}x' + \frac{4}{5}y'\right) +$$

$$+ 11\left(\frac{3}{5}x' + \frac{4}{5}y'\right)^2 = 50$$

$$x'^2\left(18 \cdot \frac{16}{25} + 24 \cdot \frac{12}{25} + 11 \cdot \frac{9}{25}\right) +$$

$$+ x'y'\underbrace{\left(-18 \cdot \frac{24}{25} + 24 \cdot \left(-\frac{9}{25}\right) + 24 \cdot \frac{16}{25} + 11 \cdot \frac{24}{25}\right)}_{0}$$

$$+ y'^2\left(18 \cdot \frac{9}{25} - 24 \cdot \frac{12}{25} + 11 \cdot \frac{16}{25}\right) = 50$$

$$27x'^2 + 2y'^2 = 50 \quad / : 50$$

$$\frac{x'^2}{\frac{50}{27}} + \frac{y'^2}{\frac{50}{2}} = 1$$

$$x^2 + y^2 = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\left\{ v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix},\ v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} = B$$

$$\left\{ w_1 = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix},\ w_2 = \begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix} \right\} = B'$$

$$w_1 = \alpha_{11} v_1 + \alpha_{21} v_2 \iff \begin{cases} \frac{4}{5} = \alpha_{11} \cdot 1 + \alpha_{21} \cdot 1 \\ \frac{3}{5} = 2\alpha_{11} + \alpha_{21} \end{cases}$$

$$\Rightarrow \alpha_{11} = -\frac{1}{5},\ \alpha_{21} = 1$$

$$w_2 = \alpha_{12} v_1 + \alpha_{22} v_2$$

$$\begin{pmatrix} -\frac{1}{5} & \frac{7}{5} \\ 1 & -\frac{10}{5} \end{pmatrix} = P_B^{B'}$$

$$\iff \begin{cases} -3/5 = \alpha_{12} + \alpha_{22} \\ 4/5 = 2\alpha_{12} + \alpha_{22} \end{cases} \Rightarrow \alpha_{12} = \frac{7}{5}$$

$$\alpha_{22} = -\frac{10}{5}$$